Muhammad Subhan Rauf

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About Me

- Computer Science student at FAST National University (graduating May 2026, with 'A's in core programming/math), pursuing knowledge and practical experience in Large Language Models (LLMs), Generative AI, and Machine Learning.

Education

FAST National University of Computer and Emerging Sciences
Bachelor's in Computer Science (BSCS) — Peshawar, Pakistan
CGPA: 3.2/4.0 — August 2022-present

F.Sc Divisional Public School and College
Pre-Engineering — Sahiwal, Pakistan
76.3% — July 2020-June 2022

Matric Divisional Public School and College
Science Subjects — Sahiwal, Pakistan
92.2% — March 2018-May 2020

Academic Projects

YouTube Video Q&A Retrieval Augmented Generation Streamlit App

- Developed an interactive web application using Streamlit for Q&A on YouTube video transcripts.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline leveraging **Hugging Face models** for chat (Mistral-7B-Instruct-v0.2) and embeddings (all-MiniLM-L6-v2).
- Enabled users to input a YouTube URL, extract transcripts, and ask questions, providing accurate answers based on video content.
- https://github.com/Mrup1/youtube-video-Q-A-RetrivalAugmentedGeneration-streamlit

AI Model Fine-Tuning for Entry Test Preparation

- Developed and implemented a fine-tuning pipeline for the Mistral-7B-Instruct-v0.2 LLM using 4-bit quantization and LORA, leveraging PyTorch, Hugging Face Transformers, TRL, and PEFT.
- Trained the model on a custom dataset of past entry test exams, significantly enhancing its ability to generate accurate responses to exam-style instructions.
- https://github.com/Mrup1/FineTunedModel

Formula 1 Prediction AI Project

- Contributed to an academic AI project focused on predicting Formula 1 race outcomes using historical data.
- Applied foundational machine learning concepts and utilized Python with libraries such as Pandas, NumPy, and Scikit-learn for data analysis and model development.
- Demonstrated ability to work with data, analyze patterns, and approach predictive modeling problems within an academic setting.
- https://github.com/Mrup1/projects.git

Bank Management System

- Developed a console-based Bank Management System using C++.
- Implemented core OOP concepts including classes, hierarchical inheritance, and polymorphism (virtual functions).
- Managed account operations (create, deposit, withdraw, check balance, modify, close) and utilized static members for tracking.
- https://github.com/Mrup1/oop-project.git

Technical Skills

- **Programming Languages:** C, C++, Python
- **Tools:** VS Code, Jupyter Notebook, Power BI, Draw.io
- **Concepts & Interests:** Artificial Intelligence (AI), Large Language Models (LLMs), Generative AI, Machine Learning (ML), Fine-tuning, Chatbot Integration, Web Development